SETUP AND RUN GUIDE

Healthcare Intelligence on Snowflake

Build the whole platform from an empty account, step by step: data, semantic view, agent, skills, app, evaluation and tests.

Follow the steps in order. Each step says where to do it, what to run, and what you should see.

The order at a glance

Step	What	Where	Time
0	Check the account (edition, role, AI access)	Worksheet	5 min
1	Get the Marketplace data	Marketplace	5 min
2	Put the project files in a Workspace folder	Workspace	10 min
3	Check config.py	Workspace	5 min
4	Build the data layers: files 01 to 05	Notebook	20-30 min
5	Create the agent	UI (or file 06)	15 min
6	Upload and add the 3 skills	UI	10 min
7	Build the rest: files 07 to 10	Notebook	10-15 min
8	Create the Streamlit app	UI (or run.create_app)	10 min
9	Run the evaluation	Notebook + UI	10-15 min
10	Test everything	Worksheet + app	15 min

Steps 11 to 14 cover re-running, sharing with a client, troubleshooting, and costs and cleanup.

Before you start: how the files work

File	Role	You...
config.py	Every object name in one place	**edit** it, never run it
run.py	Reads each .sql file, fills in the {{NAMES}} from config.py, runs it	**run** it from a notebook
01 ... 11 .sql	The build instructions	never paste them into a worksheet (placeholders)
streamlit_app.py	The app	paste it into the Streamlit editor
skills/	3 agent skills (SKILL.md each)	upload to the skills stage
agent_eval_dataset.sql	Dataset for the UI evaluation	run it in a worksheet (no placeholders)

WATCH OUT Placeholders

The .sql files contain names such as {{DATABASE}}. Only `run.py` replaces them, using `config.py`. Pasting a .sql file straight into a worksheet gives an error.

```
# The three functions in run.py
import run
run.run()                                          # all files in SQL_FILES, in order
run.run(["09_audit_log.sql", "10_security.sql"])   # only these files
run.create_app()                                   # creates the Streamlit app from code
run.run_evaluation()                               # asks the agent the 13 test questions
```

Step 0. Check the account

Where: Snowsight worksheet, role ACCOUNTADMIN **Time:** 5 minutes

Requirement	Why	How to check
Enterprise edition or higher	Masking policies (file 02) need it	Admin > Accounts: Edition column, or create a test masking policy
Paid account (not a free trial)	Trial accounts can block AI_COMPLETE and agent features	Run the AI test below
Role **ACCOUNTADMIN**	run.py creates roles, grants and account settings	Top right of Snowsight

Run this test once:

```
USE ROLE ACCOUNTADMIN;
ALTER ACCOUNT SET CORTEX_ENABLED_CROSS_REGION = 'ANY_REGION';   -- lets Cortex use models in
    other regions
SELECT AI_COMPLETE('claude-sonnet-4-5', 'Say hello');            -- must return text, not an
    error
```

WATCH OUT If the test fails

"not available for trial accounts": add a payment method or use a paid account.

"model not available": try another model, e.g. `'mistral-large2'`, and use it as AI_MODEL in config.py.

Step 1. Get the Marketplace data

Where: Snowsight > Data Products > Marketplace **Time:** 5 minutes

1. Search for **Synthetic Healthcare Data (Clinical and Claims)**.
2. Click **Get** and accept the terms.
3. Open **Catalog > Databases** and note the exact database name. The tables are in the **SILVER** schema.

The data is a read-only share: nothing is copied and you cannot create objects inside it. Everything is built in our own database, HEALTHCARE_INTELLIGENCE.

Step 2. Put the project files in a Workspace folder

Where: Snowsight > Projects > Workspaces **Time:** 10 minutes

1. Create a folder, for example **Healthcare_Intelligence**.
2. Upload **all** files into that one folder (not in subfolders): config.py, run.py, streamlit_app.py and every .sql file from 01 to 11.
3. Keep the **skills** folder on your computer for step 6.
4. In the same folder, create a **Notebook** (any name, e.g. *build_platform*).

```
Healthcare_Intelligence/
  config.py   run.py   streamlit_app.py
  01_setup.sql  02_business_views.sql  03_derived_tables.sql  04_search.sql
  05_semantic_view.sql  06_agent.sql  07_marts.sql  08_evaluation.sql
  09_audit_log.sql  10_security.sql  11_snowflake_intelligence.sql
  build_platform (notebook)
```

REMEMBER Why the same folder

run.py looks for the .sql files and config.py next to itself. A notebook in the same folder can `import run` directly.

Step 3. Check config.py

Where: Workspace: open config.py **Time:** 5 minutes

Setting	Value	Change it when...
SOURCE / SOURCE_DATABASE	SYNTHETIC_HEALTHCARE_DATA_CLINICAL_AND_CLAIMS(.SILVER)	your Marketplace database has another name (step 1)
DATABASE / WAREHOUSE	HEALTHCARE_INTELLIGENCE / HEALTHCARE_INTELLIGENCE_WH	you want other names
ANALYTICS, SEARCH, MARTS, GOVERNANCE	schema names	rarely
Roles	DATA_ENGINEER, DATA_ANALYST, HC_EXECUTIVE, HC_CLINICAL_ANALYST, HC_FINANCE_ANALYST	rarely

Setting	Value	Change it when...
SEMANTIC_VIEW / SEARCH_SERVICE	HEALTHCARE_SEMANTIC_MODEL / CLINICAL_TERMS_SEARCH	rarely
AGENT	**HEALTHCARE_AGENT_UI** (agent made in the UI) or HEALTHCARE_AGENT (file 06)	**must match the agent you create in step 5**
APP	HEALTHCARE_APP	only used by run.create_app()
AI_MODEL	claude-sonnet-4-5	the model test in step 0 failed

Step 4. Build the data layers (files 01 to 05)

Where: Notebook in the project folder **Time:** 20-30 minutes

Run this in one notebook cell:

```
import run
run.run(["01_setup.sql", "02_business_views.sql", "03_derived_tables.sql",
         "04_search.sql", "05_semantic_view.sql"])
```

You should see:

```
[OK] 01_setup.sql
[OK] 02_business_views.sql
[OK] 03_derived_tables.sql      (the longest: 763mn measurements, MEDIUM warehouse)
[OK] 04_search.sql
[OK] 05_semantic_view.sql
```

File	Creates
01	Warehouse, database, 4 schemas, 5 roles (granted to you)
02	12 business views, 4 role-based masking policies, PATIENTS_WITH_PII function
03	PATIENT_LATEST_MEASUREMENTS, PATIENT_PROFILE, HOSPITAL_STAYS
04	CLINICAL_TERMS + Cortex Search service + AGENT_SKILLS stage
05	Semantic view HEALTHCARE_SEMANTIC_MODEL

Quick check (worksheet):

```
SELECT COUNT(*) FROM HEALTHCARE_INTELLIGENCE.ANALYTICS.PATIENT_PROFILE;     -- about 1.42
    million
SHOW SEMANTIC VIEWS IN SCHEMA HEALTHCARE_INTELLIGENCE.ANALYTICS;            --
    HEALTHCARE_SEMANTIC_MODEL
SHOW CORTEX SEARCH SERVICES IN SCHEMA HEALTHCARE_INTELLIGENCE.SEARCH;       --
    CLINICAL_TERMS_SEARCH
```

REMEMBER If a file fails

run.py prints the failing statement and stops. Fix the cause (see Step 13), then run again from that file, e.g. `run.run(["03_derived_tables.sql", "04_search.sql", "05_semantic_view.sql"])`. The files are safe to re-run.

Step 5. Create the agent

Where: Snowsight > AI & ML > Agents (option A) or notebook (option B) **Time:** 15 minutes

Option A: in the UI (what this project did; agent name **HEALTHCARE_AGENT_UI**)

1. AI & ML > **Agents** > **Create agent**. Database **HEALTHCARE_INTELLIGENCE**, schema **ANALYTICS**, name **HEALTHCARE_AGENT_UI** (the same as AGENT in config.py).
2. **Overview / About:** display name *Healthcare Intelligence Assistant*; a one-line description; add the 5 sample questions (list below).
3. **Orchestration model:** auto. **Budget:** 120 seconds, 20,000 tokens.
4. **Instructions:** paste the orchestration and response text from the appendix at the end of this guide.
5. **Tools > Cortex Search:** name **clinical_term_lookup**; service **SEARCH.CLINICAL_TERMS_SEARCH**; ID and title column **CLINICAL_TERM**; max results 8; paste its description from the appendix.
6. **Tools > Cortex Analyst:** name **healthcare_analyst**; semantic view **ANALYTICS.HEALTHCARE_SEMANTIC_MODEL**; warehouse **HEALTHCARE_INTELLIGENCE_WH**; query timeout 90 s; paste its description from the appendix.
7. **Access:** leave ACCOUNTADMIN; file 10 grants USAGE to DATA_ANALYST and MONITOR to DATA_ENGINEER.
8. Click **Save**, then **Publish**. Test it in the **Preview** tab: "*How many patients have had a heart attack?*" should give 61,820.

5 sample questions:

- What is the 30-day readmission rate by age group?
- How many living adults are in each BMI category, and how common is high blood pressure in each?
- How many patients have had a heart attack, and how many of them are currently on blood pressure medication?
- How many patients received a flu vaccine each year?
- Which insurers have the most open claims?

Option B: in SQL (agent as code). Set `AGENT = "HEALTHCARE_AGENT"` in config.py, then:

```
run.run(["06_agent.sql"])
```

WATCH OUT Pick one way only

Never run file 06 with AGENT set to your UI agent's name: `CREATE OR REPLACE AGENT` would overwrite the UI agent, including its skills and CoWork setting.

If you use option A, comment out `"06_agent.sql"` and `"11_snowflake_intelligence.sql"` in SQL_FILES in run.py, so `run.run()` can never replace it.

Step 6. Upload and add the 3 skills

Where: Snowsight: Catalog (stage) and the agent's Skills tab **Time:** 10 minutes

1. Catalog > HEALTHCARE_INTELLIGENCE > ANALYTICS > Stages > **AGENT_SKILLS** (created by file 04) > **+ Files**.
2. Upload so the stage has exactly these 3 paths:

```
skills/number_formatting/SKILL.md
skills/chart_types/SKILL.md
skills/chart_styling/SKILL.md
```

1. Check in a worksheet: `LIST @HEALTHCARE_INTELLIGENCE.ANALYTICS.AGENT_SKILLS;` shows 3 files.
2. Agent > **Edit** > **Skills** > **Add skill**, once for each of the 3 folders.
3. **Save** and **Publish**.
4. Test in Preview: "*How many patients are there?*" should answer **1.42mn (1,421,656)**.

WATCH OUT Common mistakes

The file must be called exactly **SKILL.md**, one per folder (not SKILL_style.md).

The stage must use server-side encryption (SNOWFLAKE_SSE). File 04 already creates it that way.

Never use `CREATE OR REPLACE STAGE` on this stage: it deletes the uploaded skills.

Step 7. Build the rest (files 07 to 10)

Where: Notebook **Time:** 10-15 minutes

```
import importlib, config, run
importlib.reload(config); importlib.reload(run)      # picks up any edits to config.py /
    run.py
run.run(["07_marts.sql", "08_evaluation.sql", "09_audit_log.sql", "10_security.sql"])
```

File	Creates
07	8 KPI marts from the semantic view, monthly and clustered (MEDIUM warehouse, then back to XSMALL)
08	EVAL_GOLD_QUESTIONS (expected answers calculated by SQL), RUN_AGENT_EVAL procedure, EVAL_SCORECARD view
09	AGENT_EVENTS_RAW, daily task LOAD_AGENT_EVENTS (06:00 London), AGENT_CONVERSATION_HISTORY view
10	All grants, APP_ROLE_PERMISSIONS (tabs, PII levels), APP_USER_ROLES, app log and feedback tables

IN OUR PROJECT Snowflake CoWork (optional)

To also chat with the agent in Snowflake's own chat screen, click **Add to Snowflake CoWork** on the agent page. File 11 does the same in SQL, but the button is simpler.

Step 8. Create the Streamlit app

Where: Snowsight > Projects > Streamlit (option A) or notebook (option B) **Time:** 10 minutes

Option A: in the UI (what this project did)

1. Projects > **Streamlit** > **+ Streamlit App**.
2. Title: *Healthcare Intelligence*. Location: **HEALTHCARE_INTELLIGENCE.ANALYTICS**. Warehouse: **HEALTHCARE_INTELLIGENCE_WH**. Choose the **warehouse** runtime (not the container runtime).
3. In the editor, replace all code in **streamlit_app.py** with the project's streamlit_app.py.
4. Add a new file **config.py** in the app and paste the project's config.py (the app cannot see your Workspace).
5. **Packages** (top of the editor) > add **reportlab** (for the PDF report). Added once; it is saved in environment.yml.
6. Click **Run**.

Option B: from the notebook (uploads the files and adds reportlab automatically)

```
run.create_app()
```

What you should see: 9 tabs for DATA_ENGINEER (Overview, Clinical, Hospital, Pharmacy & Vaccines, Claims & Cost, Patients, AI Insights, Governance, Quality), a role dropdown, Year/Month switch, period slider and **Generate report** in the sidebar.

> **WATCH OUT** **If you see "pyproject.toml file does not exist"**
> The app was created on the container runtime. Re-create it on the warehouse runtime, or use run.create_app(), which sets `RUNTIME_NAME = 'SYSTEM$WAREHOUSE_RUNTIME'`.

Step 9. Run the evaluation

Where: Notebook (SQL harness) and the agent's Evaluations tab (UI) **Time:** 10-15 minutes

A. SQL harness (rule-based, shown in the app's Quality tab)

```
run.run_evaluation()      # asks the agent 13 questions, 5-10 minutes; returns PASS / FAIL per
    question
```

Expected result: **13 / 13 passed**. Open the app as DATA_ENGINEER > **Quality** tab to see it.

B. UI Agent Evaluations (AI judge)

1. Run **agent_eval_dataset.sql** in a worksheet. It creates GOVERNANCE.AGENT_EVAL_DATASET: input_query (VARCHAR) + ground_truth (VARIANT JSON with `ground_truth_output`).
2. Agent > **Evaluations** > **Start evaluation run** > **Run an eval manually**.
3. Dataset: **create a new dataset from a table** > AGENT_EVAL_DATASET. Map Input = INPUT_QUERY, Ground truth = GROUND_TRUTH.
4. Metrics: what is available (Logical consistency). Start.
5. Second run: **use the existing dataset**, so **Answer correctness** can be switched on as well.

6 Set the date filter to **Last 7 days** to see older runs.

If a run fails with **403 "not allowed to access this endpoint"**, the AI judge is not enabled on the account; use the SQL harness result instead, or ask Snowflake support.

Step 10. Test everything

Where: Worksheet and app **Time:** 15 minutes

Masking policies (worksheet, run each line separately)

```
USE ROLE DATA_ENGINEER;       SELECT FIRST_NAME, SOCIAL_SECURITY_NUMBER, STREET_ADDRESS,
    DATE_OF_BIRTH FROM HEALTHCARE_INTELLIGENCE.ANALYTICS.PATIENTS LIMIT 3;
USE ROLE HC_CLINICAL_ANALYST; SELECT FIRST_NAME, SOCIAL_SECURITY_NUMBER, STREET_ADDRESS,
    DATE_OF_BIRTH FROM HEALTHCARE_INTELLIGENCE.ANALYTICS.PATIENTS LIMIT 3;
USE ROLE DATA_ANALYST;        SELECT FIRST_NAME, SOCIAL_SECURITY_NUMBER, STREET_ADDRESS,
    DATE_OF_BIRTH FROM HEALTHCARE_INTELLIGENCE.ANALYTICS.PATIENTS LIMIT 3;
```

Role	Names	SSN	Address	Date of birth
DATA_ENGINEER	real	real	real	real
HC_CLINICAL_ANALYST	real	XXX-XX-1234	***MASKED***	real
DATA_ANALYST	***MASKED***	XXX-XX-XXXX	***MASKED***	year only

App checklist

Check	Expected
Switch role in the sidebar	Tabs appear and disappear; the PII caption changes
Patients tab as each role	Full / partial / masked personal data
Overview: AI summary button	3 bullet points (AI_COMPLETE)
AI Insights: question 3 from the dropdown	61.82K (61,820) and 42.16K (42,161), chart or table, SQL in the expander
AI Insights: a question with "John Smith, SSN 123-45-6789"	Yellow warning: identifiers removed (AI_REDACT), then a refusal
Feedback box after an answer	"Sentiment (AI_SENTIMENT): positive"
Governance tab	Questions by topic, feedback chart, audit log (click Refresh audit log)
Quality tab	13 / 13 passed
Generate report	Download report (PDF) with page numbers

Step 11. Re-running and day-to-day

You changed...	Then run	Why
config.py	The files that use the changed name	Names are filled in at run time
02 (business views / masking)	02, then 10	Re-created objects lose their grants
03 (derived tables)	03, then 07 and 10	Marts are built from the data
05 (semantic view)	05, then 07 and 10	Marts come from the semantic view; grants are removed on replace
The semantic view in the UI	Copy the change into file 05 (do not re-run 05 first)	Otherwise 05 overwrites the UI change
The agent (UI)	Save + Publish; never run 06	06 would replace the UI agent
08 (evaluation)	08, then 10, then run.run_evaluation()	New tables need grants
streamlit_app.py	Paste into the app editor > Run	Packages stay installed

Audit log: the task LOAD_AGENT_EVENTS runs every day at 06:00 London time. To load new questions now, click **Refresh audit log** in the Governance tab, or run `EXECUTE TASK HEALTHCARE_INTELLIGENCE.GOVERNANCE.LOAD_AGENT_EVENTS;`

Step 12. Share the app with a client

```
USE ROLE ACCOUNTADMIN;
CREATE USER CLIENT_DEMO PASSWORD = 'ChangeMe_2026!' MUST_CHANGE_PASSWORD = TRUE
  DEFAULT_ROLE = HC_EXECUTIVE DEFAULT_WAREHOUSE = HEALTHCARE_INTELLIGENCE_WH;
GRANT ROLE HC_EXECUTIVE TO USER CLIENT_DEMO;
GRANT USAGE ON STREAMLIT HEALTHCARE_INTELLIGENCE.ANALYTICS.<your_app_name> TO ROLE
    HC_EXECUTIVE;
INSERT INTO HEALTHCARE_INTELLIGENCE.GOVERNANCE.APP_USER_ROLES VALUES ('CLIENT_DEMO',
    'HC_EXECUTIVE');
```

Then open the app > **Share** > copy the link and send it with the login details. Never give a client DATA_ENGINEER. To remove access: `ALTER USER CLIENT_DEMO SET DISABLED = TRUE;`

Step 13. Troubleshooting

Message or symptom	Cause	Fix
Unsupported feature 'MASKING POLICY'	Standard edition	Use an Enterprise account (or IFF masking in a secure view)
... is not available for trial accounts	Free trial	Add a payment method / use a paid account
Insufficient privileges on SYNTHETIC_...	Marketplace share is read-only	Create objects only in HEALTHCARE_INTELLIGENCE
Client-side encryption stages are not supported	Stage without SSE	ENCRYPTION = (TYPE = 'SNOWFLAKE_SSE')
Skill does not appear / Add skill greyed	Wrong file name or folder	One folder per skill with SKILL.md; LIST the stage

Message or symptom	Cause	Fix
pyproject.toml file does not exist	Container runtime	Warehouse runtime, or run.create_app()
Download report gives HTML, not PDF	reportlab not added	App editor > Packages > reportlab
"You have no role for this app"	No row in APP_USER_ROLES	INSERT the user and role (step 12)
Analyst gets 'does not exist or not authorized'	Object re-created, grants lost	run.run(["10_security.sql"])
Agent shows 1,421,656 not 1.42mn	Instructions conflict with the skill	Use the response text in the appendix; Publish
Evaluation: Failed to validate GROUND_TRUTH	Column is OBJECT	Cast to ::VARIANT (agent_eval_dataset.sql does)
Evaluation: 403 not allowed	AI judge not enabled	Use run.run_evaluation(); ask support
Governance tab empty	Audit log not loaded yet	Refresh audit log; wait about a minute
A run.py file fails half-way	One statement failed	Read the printed statement, fix, re-run from that file

Step 14. Costs and cleanup

Item	What it costs	Tip
Warehouse	Credits while running	XSMALL + AUTO_SUSPEND 60 s (file 01); 03 and 07 resize to MEDIUM only while building
Cortex Search	Refresh + serving	TARGET_LAG 1 day is enough
Agent / AI functions	Tokens per question	Short instructions; skills load only when needed; AI_COMPLETE is cached
Daily audit task	A few seconds a day	ALTER TASK ... SUSPEND when not needed

Remove everything (only when you are finished):

```
USE ROLE ACCOUNTADMIN;
DROP DATABASE IF EXISTS HEALTHCARE_INTELLIGENCE;     -- views, tables, semantic view, agent,
    app, search, tasks
DROP WAREHOUSE IF EXISTS HEALTHCARE_INTELLIGENCE_WH;
DROP ROLE IF EXISTS HC_EXECUTIVE; DROP ROLE IF EXISTS HC_CLINICAL_ANALYST; DROP ROLE IF EXISTS
    HC_FINANCE_ANALYST;
DROP ROLE IF EXISTS DATA_ANALYST; DROP ROLE IF EXISTS DATA_ENGINEER;
```

Appendix: agent text to paste in the UI

This is the same text as in file 06_agent.sql.

Orchestration instructions

```
You are a healthcare analytics assistant for a SYNTHETIC dataset (Synthea): patients and their
latest clinical measurements, visits, diagnoses, prescriptions, procedures, vaccinations,
allergies, billing claims, hospital admissions, clinicians, facilities and insurers.

Tool rules:
0. OFFICIAL DEFINITIONS come first. These concepts are defined once in the semantic model.
   Do NOT search for them and do NOT choose terms yourself; ask healthcare_analyst to use
       them,
   and say "official definition" in the answer:
   - heart attack / MI                -> diagnoses.is_heart_attack
   - blood pressure medication        -> prescriptions.is_bp_medication
   - flu vaccine / COVID vaccine      -> vaccinations.is_flu_vaccine / is_covid_vaccine
   - 30-day readmission               -> hospital_stays.is_readmitted / readmission_rate
   - BMI, blood pressure, HbA1c level -> patients.bmi_category / blood_pressure_category /
       hba1c_category
   - open (unpaid) claims             -> claims.is_open_claim / open_claim_count
1. For any OTHER condition, medication, procedure, vaccine, allergen or visit/admission reason
    in
   everyday words (for example "sugar diabetes", "peanut allergy", "knee replacement"), FIRST
   call clinical_term_lookup to find the exact terms stored in the data.
   - Keep only results whose TERM_TYPE fits the question.
   - Include ALL clinically equivalent terms; exclude related but different terms.
2. Then call healthcare_analyst for every number, count, rate, trend, ranking or cost. Pass
    the
   exact clinical terms you selected, word for word, inside the question you send to it.
3. Never answer with numbers from memory or estimates. Every number must come from
    healthcare_analyst.
4. "Patients who have/had X" means diagnoses. "Visits for X" means visit reasons.
   "Admissions" or "hospitalizations" means hospital_stays. If unclear, choose the most likely
   meaning and say which.
5. "Is on", "currently", "active" means ongoing records (no end date) for LIVING patients
    only.
   "Ever", "has had", "history of" includes all records. If unclear, use the current meaning
       and say so.
6. Clinical measurements (BMI, blood pressure, HbA1c, glucose, cholesterol, LDL, eGFR,
    smoking)
   are the MOST RECENT value per patient. There is no measurement history: if asked how a
   measurement changed over time, say this data holds only the latest value, and offer the
   current distribution instead.
7. BMI categories use adult thresholds. For BMI questions, use adults (18+) unless the user
    asks
   otherwise, and say so.
8. Claim dollar balances are all 0 in this dataset. For "unpaid" or "owed" questions, answer
   with open claim counts and rates, and explain why there is no dollar amount.
9. NO_INSURANCE means uninsured patients, not an insurance company. Say so when it appears.
10. Never retrieve, list or guess personal identifiers (names, SSNs, addresses, birth dates)
    or
    individual patient records. Offer an aggregated answer instead.
11. If the question cannot be answered from this data, say so plainly instead of guessing.
```

Response instructions

```
- Lead with the answer (the key number) in one sentence, in plain English.
- Then state which definitions, clinical terms, filters and time period were used.
- Format numbers with the number_formatting skill: counts as K / mn / bn with 2 decimals and
    the
  exact count in brackets in the first sentence (e.g. "1.42mn (1,421,656) patients"); money as
  $K / $mn / $bn; percentages with exactly 2 decimals (e.g. 17.06%).
- Draw charts with the chart_types and chart_styling skills.
- When you use an official definition, say "official definition" and name it.
- Use a small table when there are more than 3 rows.
- Keep it short: no more than about 150 words unless the user asks for detail.
- Only offer follow-ups you can do with your tools (more breakdowns, filters, comparisons).
  Never offer to schedule, email, export, save or refresh anything.
- End with one line: "Based on synthetic data; not medical advice."
```

Tool description: clinical_term_lookup (Cortex Search)

```
Looks up the exact clinical vocabulary stored in the data. Input: everyday or partial wording
(e.g. "sugar diabetes", "peanut allergy", "metformin", "knee replacement"). Returns matching
CLINICAL_TERM values with their TERM_TYPE (diagnosis, medication, procedure, measurement,
    vaccine,
allergen, visit_type, visit_reason). Use it BEFORE analytics for clinical concepts that have
    no
official definition. It does not return counts.
```

Tool description: healthcare_analyst (Cortex Analyst)

```
Answers analytical questions by writing and running SQL on the governed 12-table healthcare
semantic model. Covers: patient demographics and latest measurements (BMI, blood pressure,
    HbA1c,
cholesterol, kidney function, smoking) with clinical categories; visits (care setting, reason,
    year,
cost, insurer payments); diagnoses; prescriptions; procedures; vaccinations; allergies;
    billing claims
(open vs closed); hospital admissions (length of stay, 30-day readmission rate); clinicians,
    facilities
and insurers. Has official definitions for heart attack, blood pressure medication, flu and
    COVID
vaccines, readmission and open claims. Use it for all counts, rates, totals, averages, trends
    and
rankings. Contains no personal identifiers.
```